Road Crash Hotspots: A Comparative Spatial Data Mining Study

Data Preparation

Fall 2026

Loading Data & Packages

```
# Data
GDL_raw <- read.csv("../data/raw/GDL/siniestralimap_2015-2025.csv")
GDL_dict <- read.csv("../data/raw/GDL/diccionario_siniestralimap_2015_2025.csv")

# Packages
#install.packages("pacman")
library(pacman)
pacman::p_load(biostats,ggplot2,dplyr,lubridate,kableExtra)
```

Exploratory Data Analysis

The raw dataset contains **50,433 observations across 36 variables**. Every row in the dataset is representative of an individual involved in a car crash. Since the original data is in spanish, we changed the variable names to english and set the appropriate data type for each variable.

```
# Copy, rename, translate and prepare data ----

GDL_df <- GDL_raw %>%
  rename(
    record_number = consecutivo,
    crash_id = Id,
    crash_counter = contador_siniestro,
    date = fecha,
    year = anio,
    month = mes,
    day = dia,
    day_of_week = dia_sem,
    time_range = rango_hora,
    municipality = mun,
    street_1 = calle_1,
    street_2 = calle_2,
    longitude = x,
    latitude = y,
    intersection_coord = id_xy_int,
    crash_type = tipo_siniestro,
    user_condition = condicion_usuario,
    user_type = tipo_usuario,
    vehicle_involved = ibaen_atro,
```

```
  public_trans_jurisdiction = jurisdiccion_tp,
  sex = sexo,
  age_range = rango_edad,
  injury_severity = gravedad,
  outcome = consecuencia,
  is_fatality = dmuerto,
  is_injured = dherido,
  is_pedestrian = dpeaton,
  is_cyclist = dbici,
  is_motorcyclist = dmoto,
  is_other_vehicle = dotroveh,
  is_driver = dconductor,
  is_passenger = dpasajero,
  is_unknown_user = dileso,
  involved_persons = involucrados,
  year_month = mes_anio,
  injury_hierarchy = herido_jer
) %>%

# Standardize unknown values
mutate(across(where(is.character),
              ~ na_if(., "Desconocido"))) %>% # 'Desconocido' means 'Unkown'

# Convert dates
mutate(
  date = dmy(date),
  year_month = ym(year_month),
  day = as.integer(day),

  # Translate months
  month = recode(
    month,
    "Enero" = "January", "Febrero" = "February",
    "Marzo" = "March", "Abril" = "April",
    "Mayo" = "May", "Junio" = "June",
    "Julio" = "July", "Agosto" = "August",
    "Septiembre" = "September", "Octubre" = "October",
    "Noviembre" = "November", "Diciembre" = "December"
  ),

  # Translate days of the week
  day_of_week = recode(
    day_of_week,
    "Lunes" = "Monday", "Martes" = "Tuesday",
    "Miércoles" = "Wednesday", "Miercoles" = "Wednesday",
    "Jueves" = "Thursday", "Viernes" = "Friday",
    "Sábado" = "Saturday", "Sabado" = "Saturday",
    "Domingo" = "Sunday"
  ),

  # Translate crash_type
  crash_type = recode(
    crash_type,
```

```
    "Colision con objeto fijo" = "Collision with fixed object",
    "Colision con persona peatona" = "Collision with pedestrian",
    "Colision entre dos o mas vehiculos" = "Collision between two or more vehicles",
    "Otro" = "Other",
    "Caida de vehiculo" = "Fall from vehicle",
    "Caida al interior de un autobus" = "Fall inside a bus",
    "Colision con vehiculo estacionado" = "Collision with parked vehicle",
    "Derrape y/o volcadura" = "Skidding and/or rollover",
    "Caida al exterior del autobus" = "Fall from outside of a bus",
    "Colision con animal" = "Collision with animal"
  ),

  # Translate user_condition
  user_condition = recode(
    user_condition,
    "Peaton" = "Pedestrian",
    "Pasajero" = "Passenger",
    "Conductor" = "Driver"
  ),

  # Translate user_type
  user_type = recode(
    user_type,
    "Peaton" = "Pedestrian",
    "Vehiculo particular" = "Private vehicle",
    "Transporte empresarial/personal" = "Company/employee transport",
    "Transporte de carga" = "Freight transport",
    "Motociclista" = "Motorcyclist",
    "Transporte publico/pasajeros" = "Public/passenger transport",
    "Ciclista" = "Cyclist",
    "Otro vehiculo" = "Other vehicle",
    "Tren ligero" = "Light rail"
  ),

  # Translate vehicle_involved
  vehicle_involved = recode(
    vehicle_involved,
    "Otro vehiculo" = "Other vehicle",
    "Automovil" = "Car",
    "Van" = "Van",
    "Camioneta pickup/ carga ligera" = "Pickup/light-duty truck",
    "Motocicleta" = "Motorcycle",
    "Autobus" = "Bus",
    "Bicicleta" = "Bicycle",
    "Tren" = "Train",
    "Camion unitario" = "Single-unit truck",
    "Tractocamion" = "Tractor-trailer",
    "Tren ligero" = "Light rail",
    "Autobus articulado" = "Articulated bus",
    "Motocarro" = "Motorized tricycle",
    "Patin o scooter" = "Skateboard or scooter",
    "Suv o todoterreno" = "SUV or off-road vehicle",
    "Ciclomotor" = "Moped",
```

```
      "Otros vehiculos de baja potencia" = "Other low-powered vehicles"
    ),

    # Translate sex
    sex = recode(
      sex,
      "Hombre" = "Male",
      "Mujer" = "Female"
    ),

    # Translate injury_severity
    injury_severity = recode(
      injury_severity,
      "Traumatismo o lesion" = "Injury",
      "Traumatismo mortal" = "Fatal injury",
      "Sin traumatismo aparente en sitio" = "No apparent injury at scene"
    ),

    # Translate outcome
    outcome = recode(
      outcome,
      "Lesionado" = "Injured",
      "Fallecido" = "Deceased",
      "Ileso" = "Uninjured"
    ),

    # Convert categorical variables to factors
    month = factor(month, levels = month.name, ordered = TRUE),
    day_of_week = factor(
      day_of_week,
      levels = c("Monday", "Tuesday", "Wednesday",
                 "Thursday", "Friday", "Saturday", "Sunday")
    ),
    municipality = factor(municipality),
    time_range = factor(time_range),
    crash_type = factor(crash_type),
    user_condition = factor(user_condition),
    user_type = factor(user_type),
    vehicle_involved = factor(vehicle_involved),
    sex = factor(sex),
    age_range = factor(age_range),
    injury_severity = factor(injury_severity),
    outcome = factor(outcome),
    injury_hierarchy = factor(injury_hierarchy),

    # Convert dichotomous variables to factors
    across(
      c(crash_counter, is_fatality, is_injured, is_pedestrian, is_cyclist,
        is_motorcyclist, is_other_vehicle, is_driver,
        is_passenger, is_unknown_user),
      factor
    )
  )
```

Table 1. Dataset variable descriptions.

Original	Translated	Type	Description
consecutivo	**record_number**	numeric	Record number
Id	**crash_id**	string	Unique crash id
contador_siniestro	**crash_counter**	binary	First (1) or repeated occurrence (0)
anio	**year**	integer	Year
fecha	**date**	date	Date of the crash in dd/mm/yyyy format
mes	**month**	category	Month in which the crash occurred
dia	**day**	integer	Numerical day of the month
dia_sem	**day_of_week**	string	Day of the week
rango_hora	**time_range**	string	Two-hour interval in which crash occurred
mun	**municipality**	string	Municipality containing nearest intersection
calle_1	**street_1**	string	Name of the street where the crash occurred
calle_2	**street_2**	string	Nearest intersecting street
x	**longitude**	Numeric	Longitude of the nearest intersection
y	**latitude**	Numeric	Latitude of the nearest intersection
id_xy_int	**intersection_coord**	string	Coordinates of the nearest intersection
tipo_siniestro	**crash_type**	string	Type of road crash
condicion_usuario	**user_condition**	category	Role of the person involved
tipo_usuario	**user_type**	category	User classification
ibaen_atro	**vehicle_involved**	category	Type of vehicle involved in the crash
jurisdiccion_tp	**-**	-	Empty and not described in data dictionary
sexo	**sex**	Character	Sex of the person involved.
rango_edad	**age_range**	Character	Age range of the person involved
gravedad	**injury_severity**	Character	Severity of trauma resulting from the crash
consecuencia	**outcome**	Character	Injury classification
dmuerto	**is_fatality**	Integer	Indicates if person involved died (1 = yes)
dherido	**is_injured**	Integer	Was person involved injured? (1 = yes)
dpeaton	**is_pedestrian**	Integer	Was person involved a pedestrian? (1 = yes)
dbici	**is_cyclist**	Integer	Was person involved a cyclist? (1 = yes)
dmoto	**is_motorcyclist**	Integer	Was person involved a motorcyclist? (1 = yes)
dotroveh	**is_other_vehicle**	Integer	Did person use another vehicle type? (1 = yes)
dconductor	**is_driver**	Integer	Was the person involved a driver? (1 = yes)
dpasajero	**is_passenger**	Integer	Was person involved a passenger? (1 = yes)
dileso	**is_unknown_user**	Integer	Was person involved uninjured? (1 = yes)
involucrados	**involved_persons**	Integer	Indicator of count of involved persons
mes_anio	**year_month**	Character	Month-year combination
herido_jer	**injury_hierarchy**	Character	Injury hierarchy

```
str(GDL_df)
```

```
## 'data.frame':    50433 obs. of  36 variables:
##  $ record_number            : int  222249 227628 222227 222229 222230 222231 222232 222233 222234 22
##  $ crash_id                 : chr  "2015-01-01-625694-SISCAV-04:00 A 05:59" "2015-01-01-1-EHGR-20:00
##  $ crash_counter            : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
##  $ date                     : Date, format: "2015-01-01" "2015-01-01" ...
##  $ year                     : int  2015 2015 2015 2015 2015 2015 2015 2015 2015 2015 ...
##  $ month                    : Ord.factor w/ 12 levels "January"<"February"<..: 1 1 1 1 1 1 1 1 1 1 .
##  $ day                      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ day_of_week              : Factor w/ 7 levels "Monday","Tuesday",..: 4 4 4 4 4 4 4 4 4 4 ...
##  $ time_range               : Factor w/ 12 levels "00:00 a 01:59",..: 3 11 5 3 3 3 3 3 3 3 ...
```

```
##  $ municipality             : Factor w/ 8 levels "El Salto","Guadalajara",..: 7 2 3 2 2 2 2 2 2 2 ..
##  $ street_1                 : chr  "Acueducto Av Zap" "Del Ejercito Calz" "Lazaro Cardenas Calz" "Inc
##  $ street_2                 : chr  "Real Acueducto Av" "Rio Rhin" "Tlaquepaque" "Plutarco Elias Calle
##  $ longitude                : num  -103 -103 NA -103 -103 ...
##  $ latitude                 : num  20.7 20.7 NA 20.7 20.7 ...
##  $ intersection_coord       : chr  "20.712667, -103.409311" "20.653207, -103.33658" "Sin Coordenadas
##  $ crash_type               : Factor w/ 10 levels "Collision between two or more vehicles",..: 3 5 3
##  $ user_condition           : Factor w/ 3 levels "Driver","Passenger",..: 3 3 2 1 2 2 2 2 2 2 ...
##  $ user_type                : Factor w/ 9 levels "Company/employee transport",..: 7 7 8 1 1 1 1 1 1
##  $ vehicle_involved         : Factor w/ 17 levels "Articulated bus",..: 10 10 4 17 17 17 17 17 17 17
##  $ public_trans_jurisdiction: logi  NA NA NA NA NA NA ...
##  $ sex                      : Factor w/ 2 levels "Female","Male": 2 1 1 2 1 NA 1 2 1 2 ...
##  $ age_range                : Factor w/ 6 levels "0-17","18-27",..: NA 3 NA 4 NA NA NA NA NA NA ...
##  $ injury_severity          : Factor w/ 3 levels "Fatal injury",..: 2 2 2 1 2 2 2 1 2 2 ...
##  $ outcome                  : Factor w/ 3 levels "Deceased","Injured",..: 2 2 2 1 2 2 2 1 2 2 ...
##  $ is_fatality              : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 2 1 1 ...
##  $ is_injured               : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 2 1 2 2 ...
##  $ is_pedestrian            : Factor w/ 2 levels "0","1": 2 2 1 1 1 1 1 1 1 1 ...
##  $ is_cyclist               : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ is_motorcyclist          : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ is_other_vehicle         : Factor w/ 2 levels "0","1": 1 1 2 2 2 2 2 2 2 2 ...
##  $ is_driver                : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
##  $ is_passenger             : Factor w/ 2 levels "0","1": 1 1 2 1 2 2 2 2 2 2 ...
##  $ is_unknown_user          : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ involved_persons         : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ year_month               : Date, format: "2015-01-01" "2015-01-01" ...
##  $ injury_hierarchy         : Factor w/ 2 levels "Fallecido","Lesionado": 2 2 2 1 1 1 1 1 1 1 ...
```

First of all, we can delete the *jurisdiccion_tp* and *involved_persons* variables from the dataset, due to the former being completely empty and the later being all of class '1' (Fig. 1).

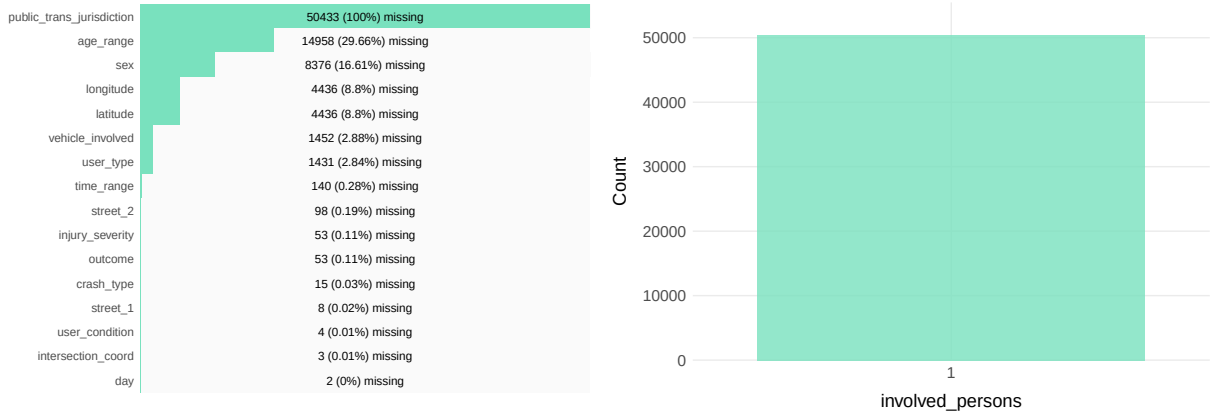


Figure 1: Distribution of missing values across variables (left) and classification of involved_persons (right).

```
GDL_df$public_trans_jurisdiction <- NULL
GDL_df$involved_persons <- NULL
```

Every row in the dataset is representative of an individual involved in a crash. We can determine the number of **unique crashes** based on the *crash_id* variable.

```r
# Unique crashes and records
GDL_df %>%
  summarise(
    observations = n(),
    unique_crashes = n_distinct(crash_id),
    missing_crash_ids = sum(is.na(crash_id))
  ) %>%
  kable()
```

Table 2. Unique crashes vs total observations.

observations	unique_crashes	missing_crash_ids
50433	18686	0